

EXPERIMENT 6

Apply K-Means and Hierarchical Clustering on Sample Datasets

Introduction to Clustering

Clustering is an **unsupervised machine learning technique** used to group similar data points together based on their characteristics.

Unlike supervised learning, clustering does not require labeled data. The goal is to discover hidden patterns or natural groupings within the dataset.

Clustering is widely used in:

- Customer segmentation
- Market analysis
- Image segmentation
- Fraud detection
- Pattern recognition

K-MEANS CLUSTERING

K-Means clustering is one of the most popular "unsupervised" machine learning algorithms. Its goal is simple: group similar data points together and discover underlying patterns without being told what to look for.

What is K-Means?

K-Means is a partition-based clustering algorithm that divides the dataset into K predefined clusters.

Each cluster is represented by its centroid (mean of points in the cluster).

The algorithm minimizes the distance between data points and their respective cluster centroids.

Working of K-Means Algorithm

1. Select number of clusters (K)
2. Randomly initialize K centroids
3. Assign each data point to nearest centroid

4. Recalculate centroids
5. Repeat until centroids stop changing

Elbow Method

The Elbow Method is used to determine the optimal value of K.

It plots:

- Number of clusters (K)
vs
- WCSS

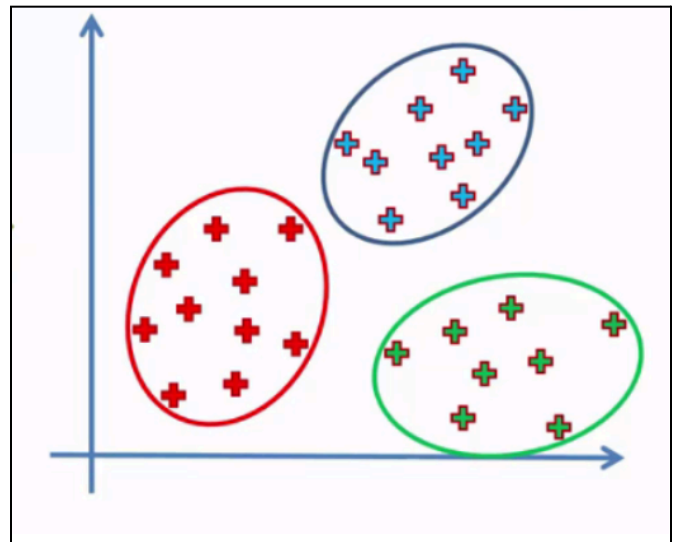
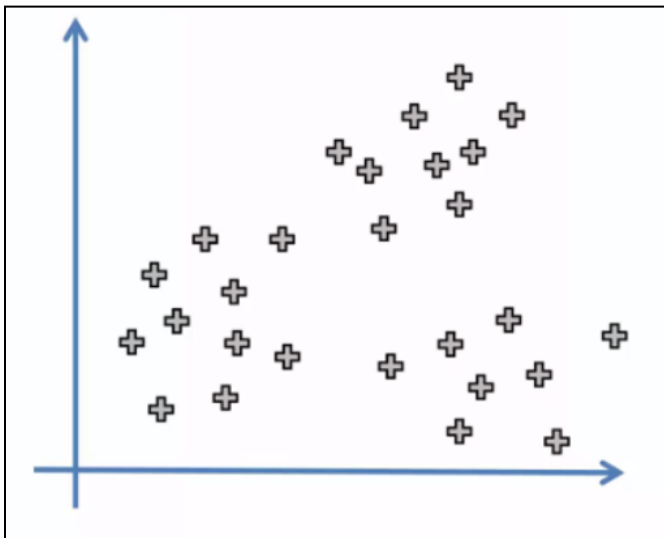
The point where the graph forms an “elbow” indicates the optimal number of clusters.

Advantages of K-Means

- Simple and easy to implement
- Efficient for large datasets
- Fast convergence

Limitations of K-Means

- Requires predefined value of K
- Sensitive to initial centroid selection
- Works poorly for non-spherical clusters
- Sensitive to outliers



HIERARCHICAL CLUSTERING

Hierarchical Clustering builds a hierarchy of clusters using a tree-like structure called a dendrogram. It does not require specifying the number of clusters initially.

There are two types:

- Agglomerative (Bottom-up approach)
- Divisive (Top-down approach)

In this experiment, Agglomerative clustering is used.

Working of Agglomerative Clustering

1. Start with each data point as its own cluster
2. Find the two closest clusters
3. Merge them
4. Repeat until all points form a single cluster

Distance Measures Used

Common linkage methods:

- Single linkage (minimum distance)
- Complete linkage (maximum distance)
- Average linkage
- Ward's method (minimizes variance)

Ward's method is commonly used because it produces compact clusters.

Dendrogram

A dendrogram is a tree diagram that shows:

- How clusters are merged
- The distance at which clusters are combined

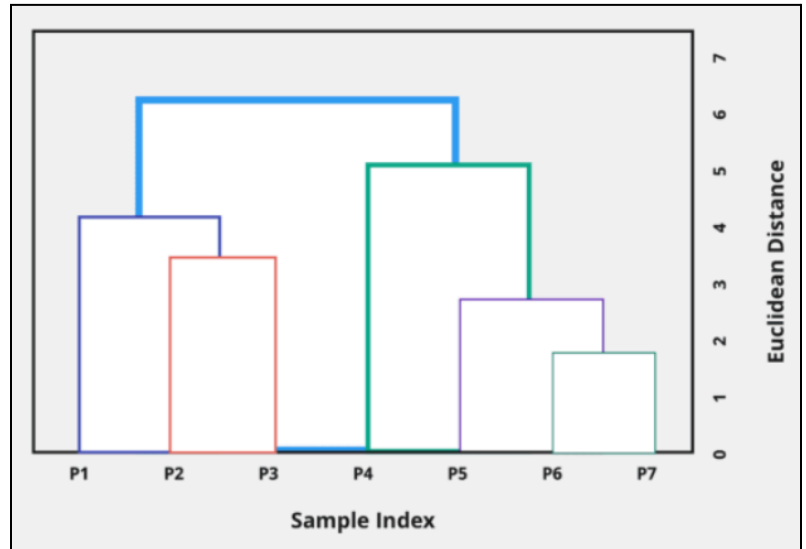
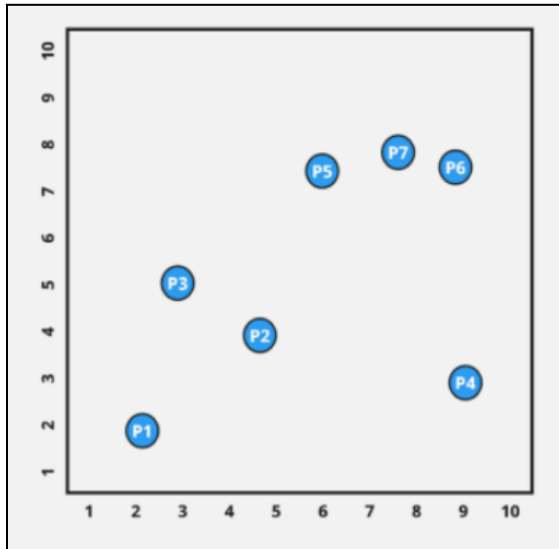
A horizontal cut across the dendrogram determines the number of clusters.

Advantages of Hierarchical Clustering

- No need to specify number of clusters initially
- Produces interpretable tree structure
- Useful for small datasets

Limitations of Hierarchical Clustering

- Computationally expensive for large datasets
- Once clusters are merged, they cannot be split
- Sensitive to noise and outliers



Comparison: K-Means vs Hierarchical Clustering

Feature	K-Means	Hierarchical
Type	Partition-based	Tree-based
Need to specify K	Yes	No
Speed	Faster	Slower
Scalability	Good for large data	Not ideal for large data
Visualization	Centroid-based	Dendrogram

1. Dataset Source

For clustering, we use synthetic dataset generated using:

`make_blobs()` from Scikit-learn

Reason:

Clustering requires unlabeled structure visualization.

2.Dataset Description

In this experiment:

- Number of samples: 25
- Number of clusters (centers): 3
- Number of features: 2
- Cluster standard deviation: 0.60

Each data point has:

- Feature 1 (X-coordinate)
- Feature 2 (Y-coordinate)

The dataset is small in size to ensure that the dendrogram is clearly visible and easy to interpret.

There is no target variable because clustering is an unsupervised learning technique.

3.Performance Analysis

Since K-Means and Hierarchical Clustering are **unsupervised algorithms**, accuracy cannot be used. So we evaluate performance using:

K-Means

- **Inertia (WCSS):** Measures how close data points are to their cluster centroid. Lower value means better clustering.
- **Elbow Method:** Used to find optimal number of clusters ($K = 3$).
- **Silhouette Score:** Shows how well clusters are separated. Value closer to 1 means better clustering.
- **Visualization:** Scatter plot showed clearly separated clusters.

Hierarchical Clustering

- **Dendrogram:** Used to decide the number of clusters. Clear separation observed at 3 clusters.
- **Silhouette Score:** Showed good cluster separation.
- **Visualization:** Final clustered graph showed distinct groups.

Both algorithms successfully formed 3 meaningful clusters.

4. Hyperparameter Tuning

Hyperparameter tuning improves clustering quality.

K-Means Tuning

- **n_clusters:** Tested multiple values using Elbow Method → Optimal $K = 3$.
- **init method:** Used *K-means++* for better centroid selection.
- **max_iter & n_init:** Increased for better convergence.

Impact: Reduced inertia and improved cluster separation.

Hierarchical Clustering Tuning

- **n_clusters:** Selected 3 based on dendrogram.
- **linkage method:** Ward method used (minimizes variance).

Impact: Produced compact and well-separated clusters.